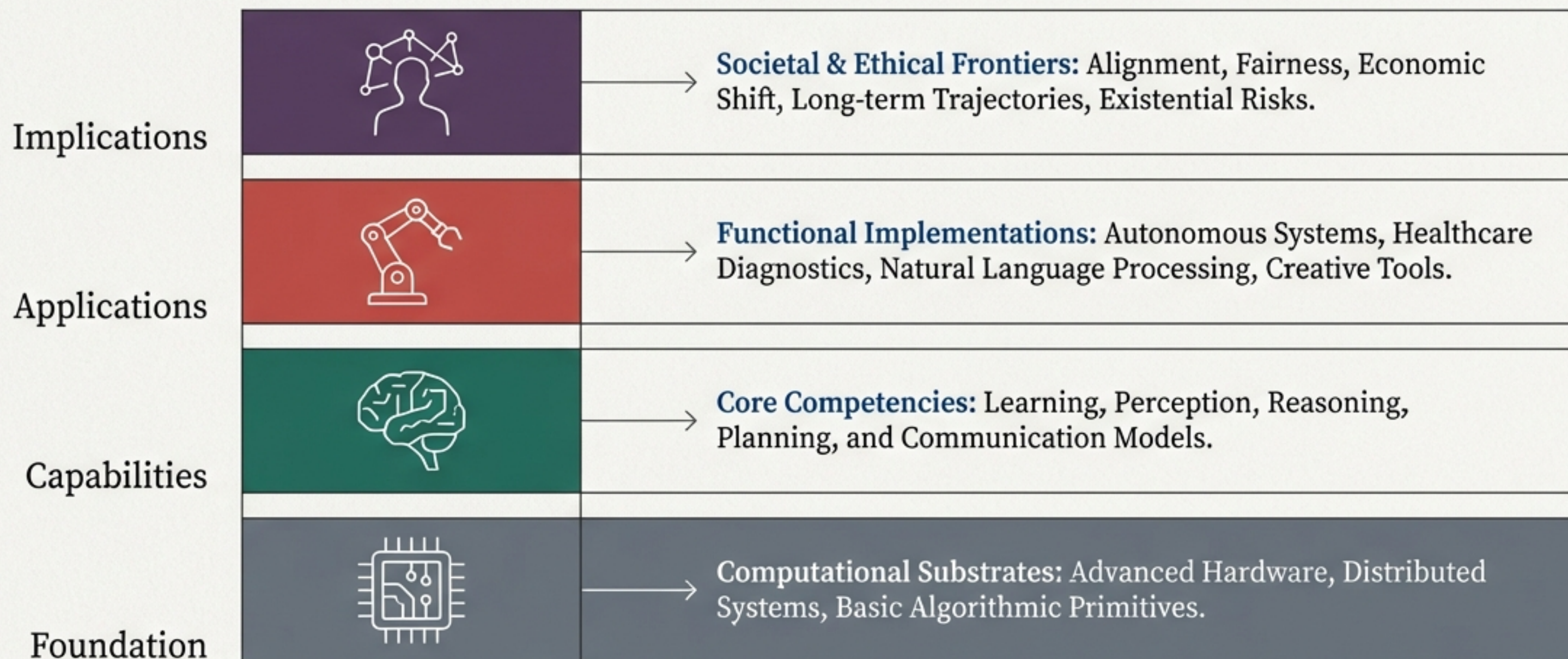


The Emergence of Intelligent Systems: A Journey Up the AI Stack

An analysis of foundational research, emergent capabilities, and frontier challenges in Artificial Intelligence.



We will ascend through a four-layer 'Capabilities Stack' to understand how AI is building upon itself.



The field of AI is advancing on multiple fronts simultaneously. To make sense of this progress, we will organize our journey through a cohesive mental model:

4. The Implications

The frontier where these powerful systems meet human society, posing the most complex technical and ethical challenges.

3. The Capabilities

The emergent cognitive abilities that arise from this foundation, focusing on the shift from raw potential to reliable reasoning.

3. The Applications

The assembly of these capabilities into autonomous systems that act in the world—the agentic leap.

1. The Foundation

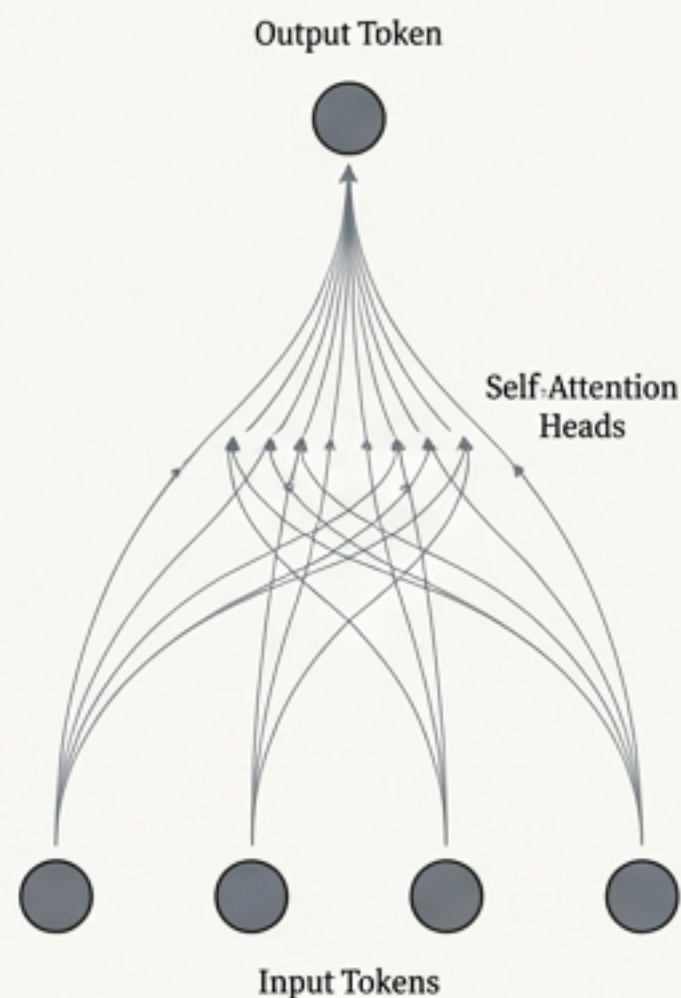
The frontier where these powerful systems meet human society, posing the most complex technical and ethical challenges.

The modern AI stack is built on the Transformer's attention mechanism and powerful generative architectures.

The Attention Revolution

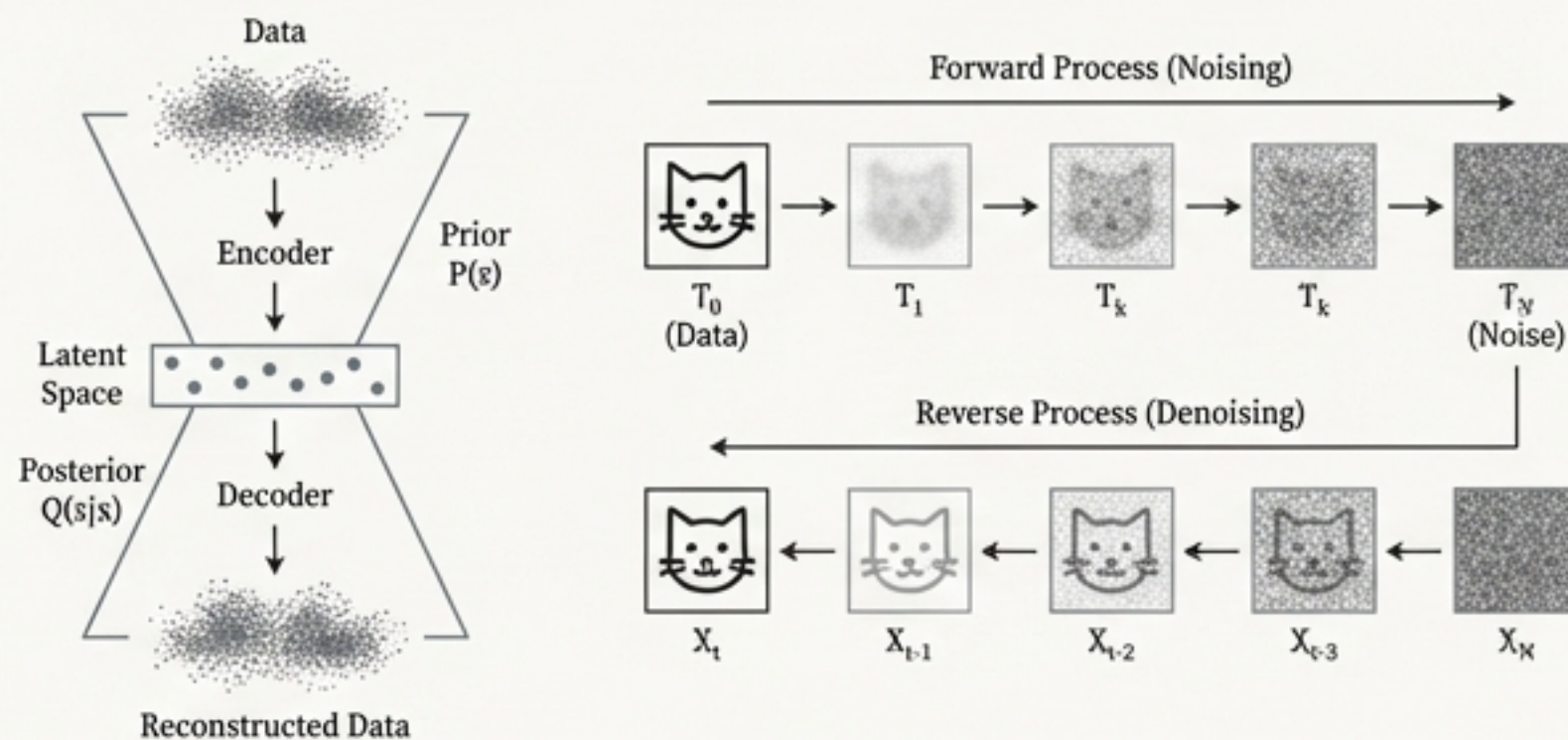
The Transformer architecture eschewed recurrence, relying entirely on self-attention to draw global dependencies. This unlocked massive parallelization and a new state-of-the-art in modeling sequences.

The original Transformer (big) achieved a BLEU score of **28.4** on WMT English-to-German, surpassing previous SOTA by over 2.0 BLEU at a fraction of the training cost (**2.3e19 FLOPs**).



The Engines of Creation

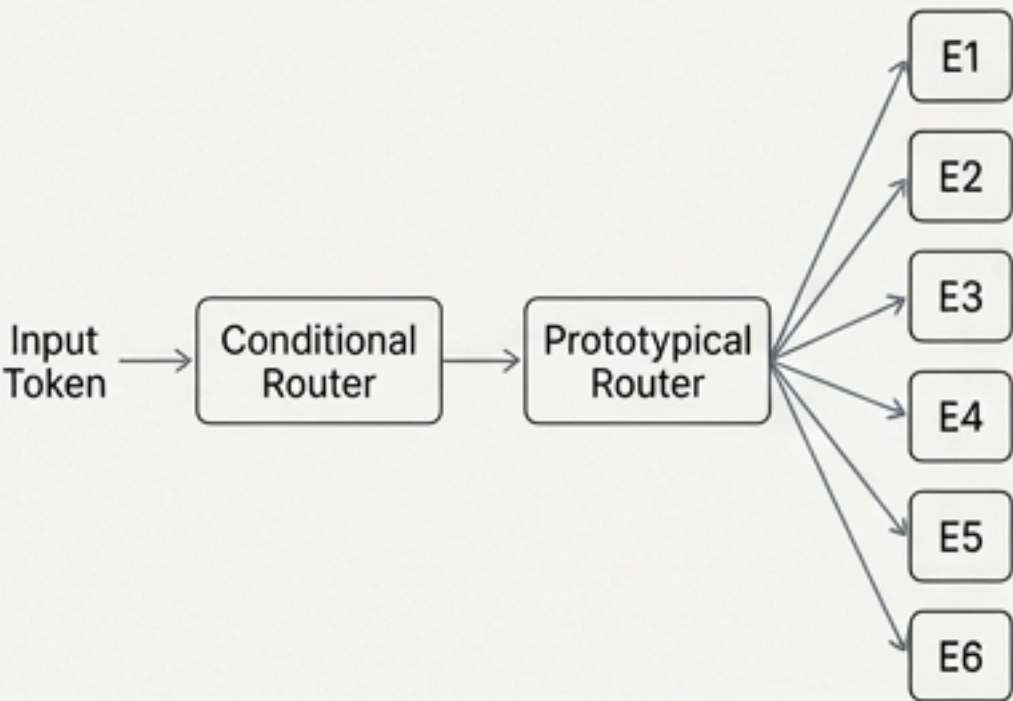
Complementing attention, Variational Autoencoders (VAEs) and Denoising Diffusion Models (DDPMs) provide the theoretical backbone for modern generative AI, learning to map from simple latent priors to complex data distributions.



Architectural innovation is relentlessly driving efficiency, achieving superior performance with less.

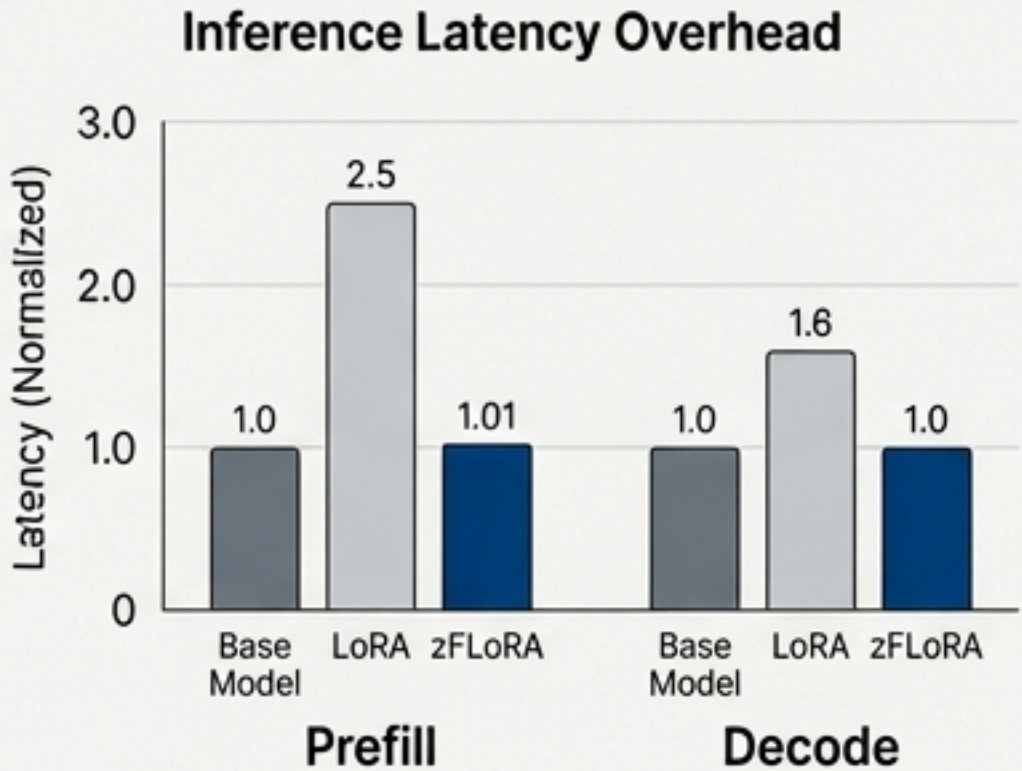
Efficient Experts (ProMoE)

ProMoE improves Mixture-of-Experts for Diffusion-in-Transformer (DiT) models with a two-step router (conditional & prototypical) that promotes true expert specialization, addressing the high spatial redundancy of visual tokens.



Zero-Latency Adapters (zFLoRA)

Standard LoRA adapters incur significant inference latency (up to 2.5x prefill, 1.6x decode). zFLoRA is a fused low-rank adapter architecture that introduces zero or negligible latency overhead while maintaining performance.



Tiny, Powerful Recursion (TRM)

Tiny Recursion Models (TRMs) demonstrate a 'less is more' principle. By decreasing layers but scaling up recursions, TRMs achieve better generalization than more complex Hierarchical Reasoning Models (HRMs) with far fewer parameters.

Performance on Sudoku-Extreme		
Model	Parameters	Accuracy
HRM	27M	55.0%
TRM	5M	87.4%

Part 2: Capabilities The Dawn of Digital Cognition

The new frontier of reasoning is formal verification, moving beyond plausible text to provable correctness.

Focus: Ax-Prover

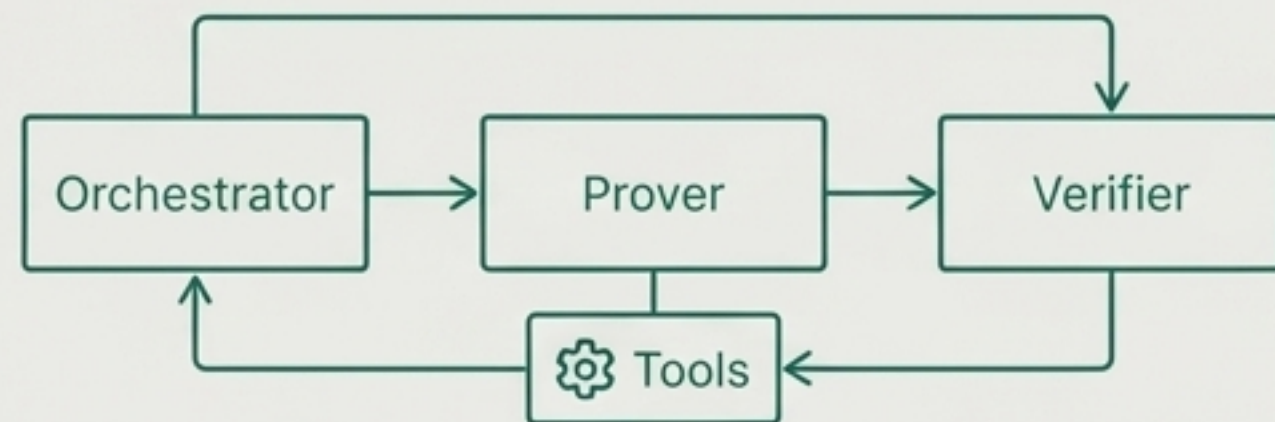
A multi-agent system (Orchestrator, Prover, Verifier) that leverages frontier LLMs for interactive theorem proving in Lean.

It sidesteps the need for specialized models and hardware by using tool-enhanced frontier LLMs, making formal verification more accessible to researchers.

Ax-Prover was evaluated not just on competition math but on two new, more abstract datasets:

- **AbstractAlgebra:** Graduate-level problems from standard textbooks.
- **QuantumTheorems:** Formal reasoning in quantum mechanics, bridging theoretical physics and formal verification.

The performance gap on these new datasets underscores the potential of this approach to act as a key AI verification tool for mathematically grounded scientific reasoning.



```
-- From the AbstractAlgebra dataset

variable {n : Nat} {i : Int}
local notation "D_n" => DihedralGroup n
local notation "r" => DihedralGroup.r (1 : ZMod n)
local notation "s" => DihedralGroup.sr (0 : ZMod n)

/-- Use the generators and relations to show that
    every element of D_n not a power of r has order 2. -/
theorem exercise_3_part1 {x : D_n} (h : x = s * r ^ i) :
  orderOf x = 2 := by sorry
```


Models are now learning to self-evolve past their limits and provide statistical certificates of their own reasoning.

Self-Evolving to Solve the Unsolvable (DSER)

Deep Self-Evolving Reasoning (DSER) is an iterative verification and refinement process that allows a model to solve problems initially beyond its baseline capacity, even with imperfect self-verification.

A powerful 8B reasoning model (DS-8B) failed to solve 9 AIME 2024-2025 competition problems.

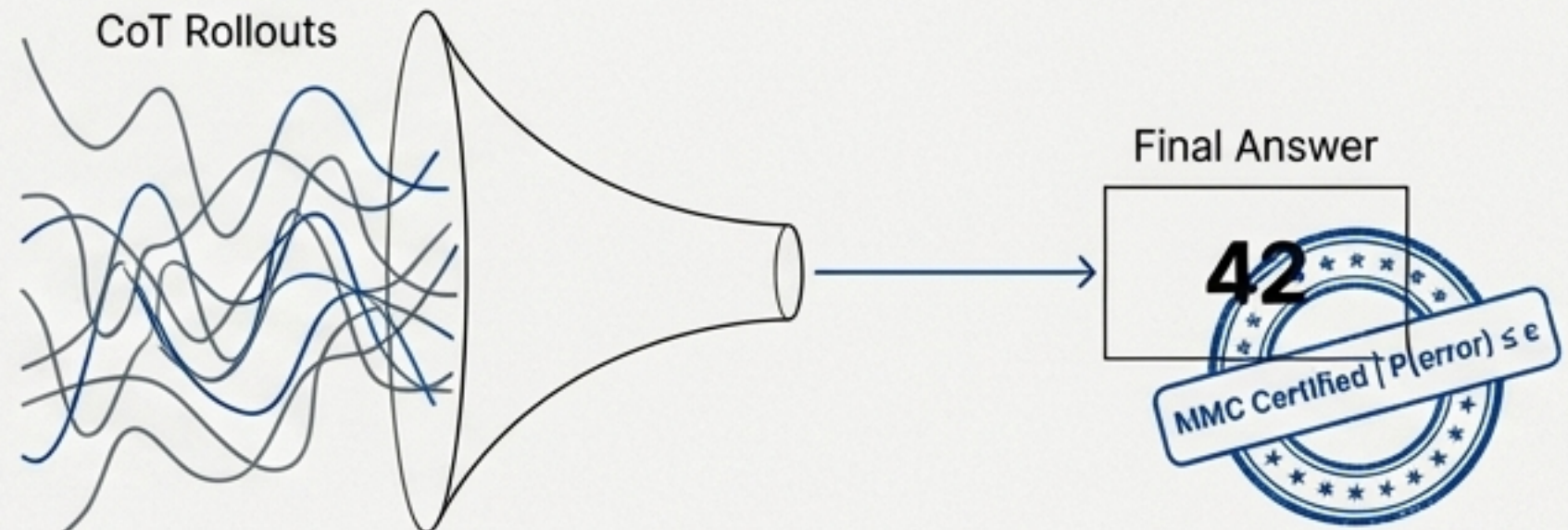


DSER's design avoids premature exits or false-positive acceptances that plague simpler self-correction methods.

Certifying Self-Consistency

We can frame Chain-of-Thought (CoT) reasoning as a statistical process. Majority voting over multiple rollouts is an estimate of the model's most probable ('modal') answer.

The Martingale Majority Certificate (MMC) is a sequential test that adaptively determines when enough rollouts have been drawn to guarantee the majority answer is the true mode, with a pre-specified error bound (e.g., $\epsilon = 0.01$).



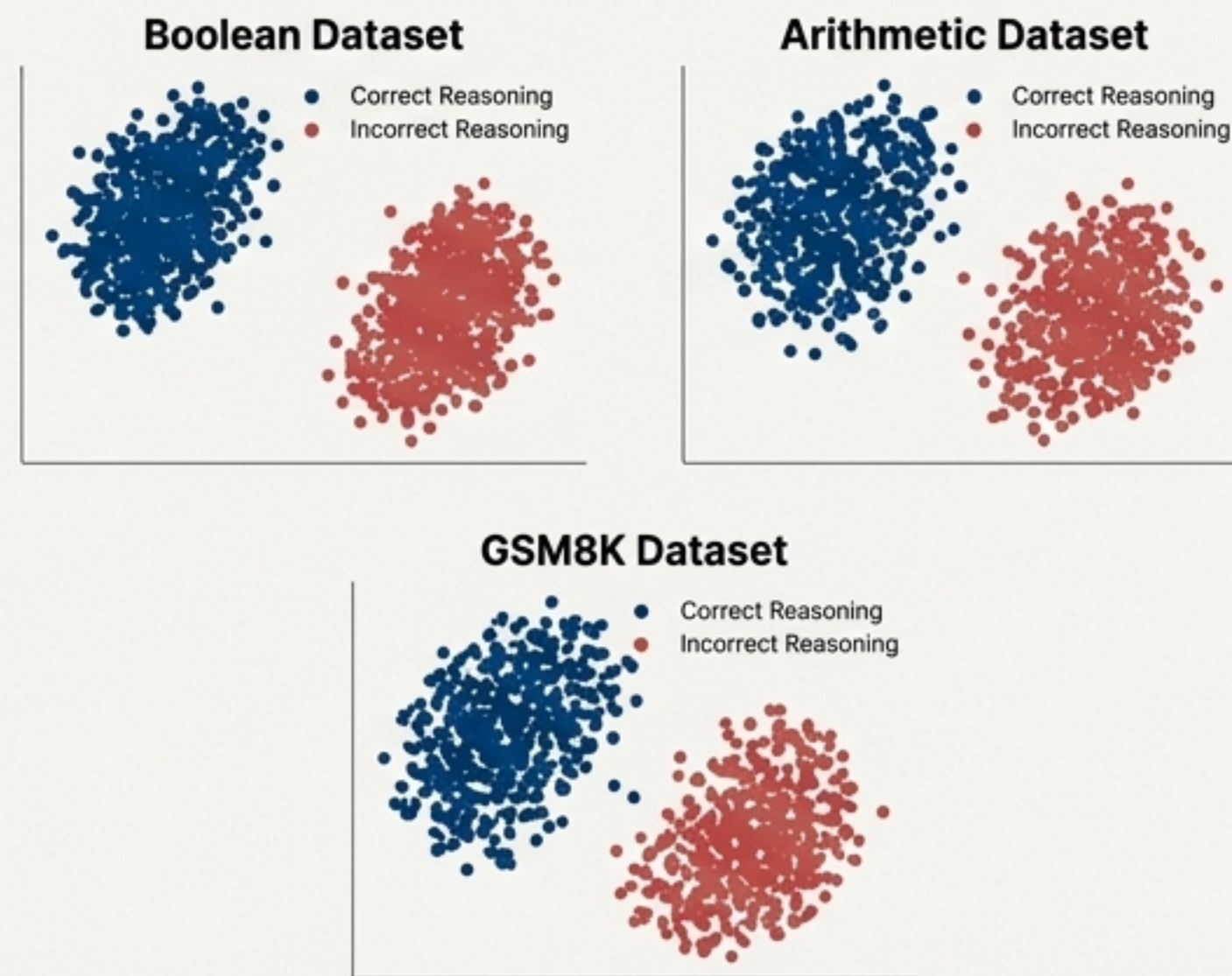
Circuit-based Reasoning Verification opens the black box to find the structural “fingerprints of error.”

Concept: Circuit-based Reasoning Verification (CRV)

CRV is a white-box methodology that analyzes the computational graph of a model during a reasoning step. It treats the attribution graph as an “execution trace” of a latent circuit. The core finding is that correct and incorrect reasoning steps leave distinct structural fingerprints.

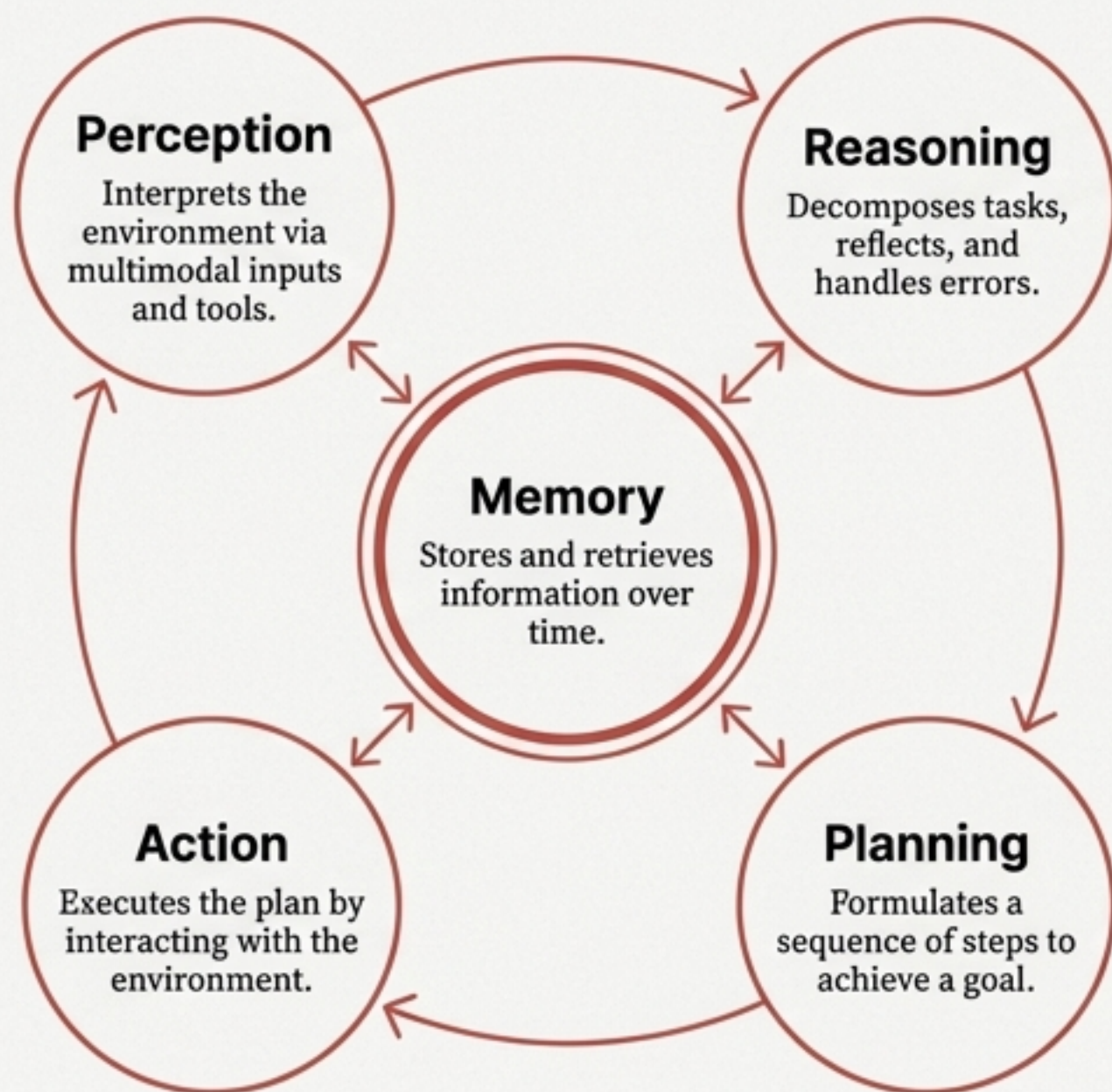
1. **Separable Signal:** The structural signatures of error are highly separable. A simple classifier trained on graph features can detect incorrect reasoning steps with high accuracy.
2. **Domain-Specificity:** Failures in different reasoning tasks (e.g., Arithmetic vs. Boolean logic) manifest as distinct computational patterns.
3. **Causal Role:** These signatures are not just correlational. Targeted interventions—amplifying or suppressing specific transcoder features—can successfully correct faulty reasoning.

t-SNE Projection of Reasoning Step Features



The clear visual separation of feature distributions validates that CRV captures a distinct structural fingerprint of error across different reasoning domains.

The Agentic Leap: Assembling cognitive capabilities into systems that perceive, remember, and act.



An intelligent agent is more than just an LLM. It is a system built upon specialized components that mimic cognitive processes to perform complex, multi-step tasks autonomously.

A comprehensive review of recent literature reveals a canonical agent architecture:

Perception: How the agent interprets its environment, using multimodal inputs, structured data (like HTML), or tool-based APIs.

Memory: How the agent stores and retrieves information over time, from short-term working memory to long-term knowledge bases.

Reasoning: How the agent decomposes tasks, reflects on its performance, and handles errors.

Planning: How the agent formulates a sequence of steps to achieve a goal.

Action: How the agent executes its plan by interacting with its environment, often through tool use.

Advanced agents are engineered with evolving playbooks, sophisticated memory, and armies of specialists.

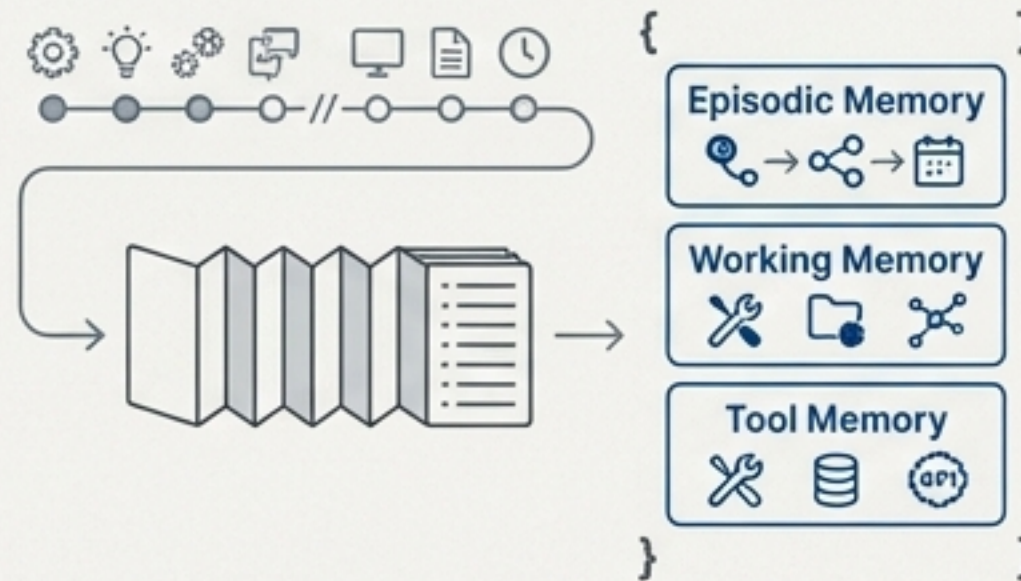
Evolving Playbooks with ACE



Agentic Context Engineering (ACE) treats context not as a summary to be compressed, but as an evolving “playbook” that accumulates and organizes strategies over time using a grow-and-refine principle.

ACE boosts accuracy on the AppWorld benchmark by **17.1%**, allowing a smaller, open-source model to match the performance of the top-ranked proprietary agent on the leaderboard.

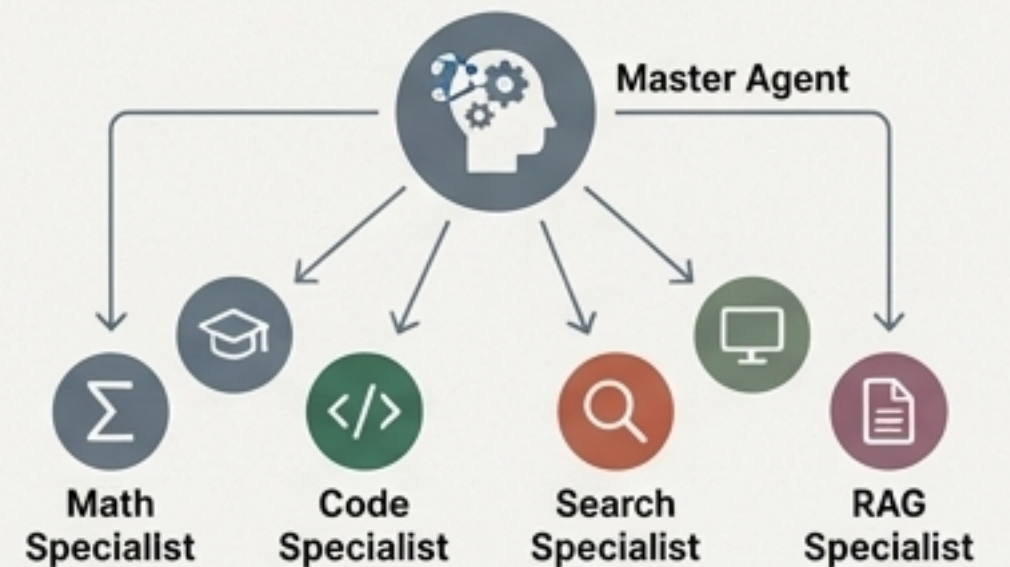
Long-Horizon Memory Folding with DeepAgent



To overcome finite context windows in long tasks, DeepAgent uses “memory folding” to compress task history into a structured JSON schema with three components: Episodic Memory, Working Memory, and Tool Memory.

Enables stable reasoning across arbitrary exploration depths without context overflow.

Generating Specialists with Alita-G



Alita-G uses a “master agent” to generate a diverse population of domain-specialist agents. It harvests Meta-Cognitive Processes (MCPs) from the master’s attempts, then filters and abstracts them to create specialized toolkits and instruction sets.

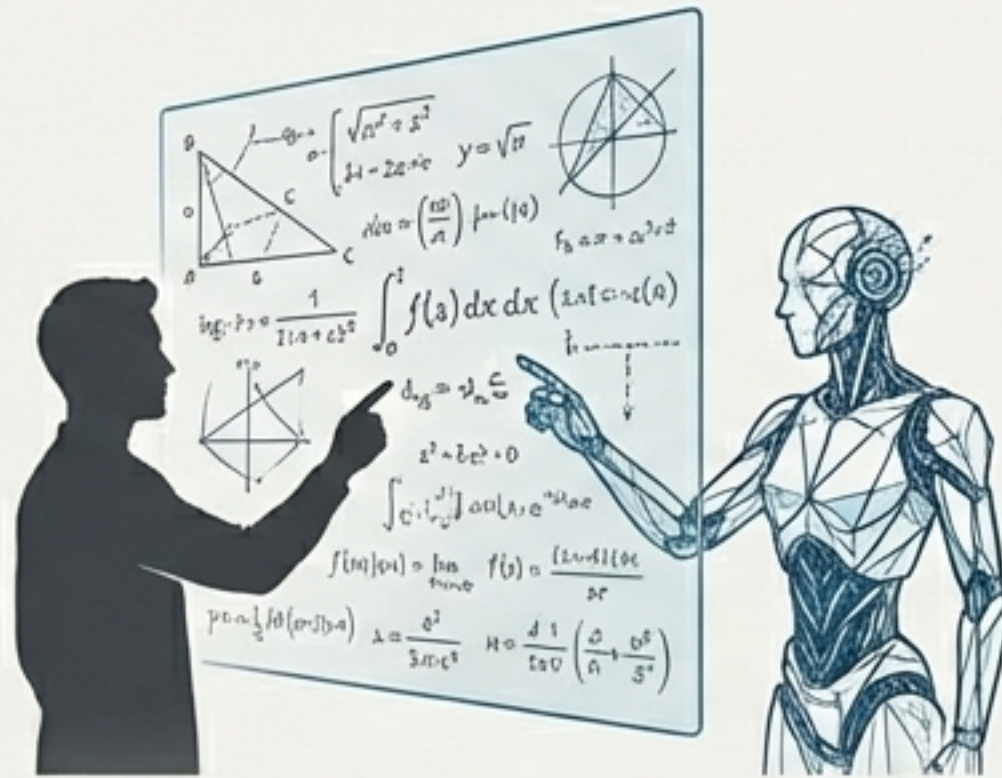
Achieves state-of-the-art **83.03% pass@1** accuracy on the challenging GAIA benchmark.

Agents in action reveal a split between powerful analytical co-reasoning and fragile real-time interaction.

Case Study 1: AI as a Mathematical Co-Reasoner (AIM)

Tackling a complex problem in homogenization theory.

A human expert collaborates with the AIM system. The human decomposes the problem, suggests high-level theoretical tools (e.g., Schauder Theory), and validates intermediate steps. AIM explores approaches, performs complex derivations, and even surfaces key properties the human expert initially missed.



“This experience vividly illustrates that although AIM may still be a flawed individual researcher today, it can already serve as a valuable research partner—if used wisely.”

Case Study 2: An Agent Playing Web Games (ChatGPT Atlas)

Playing a variety of browser games to test web interaction capabilities.

A clear performance dichotomy:

		9			
3				6	
		7		1	3
		5	8		
9	6		4		
					5

Success in Analytical Games

Solved medium-difficulty Sudoku in **2m 28s**, roughly 75% faster than the human baseline (10-12m).



Failure in Motor-Skill Games

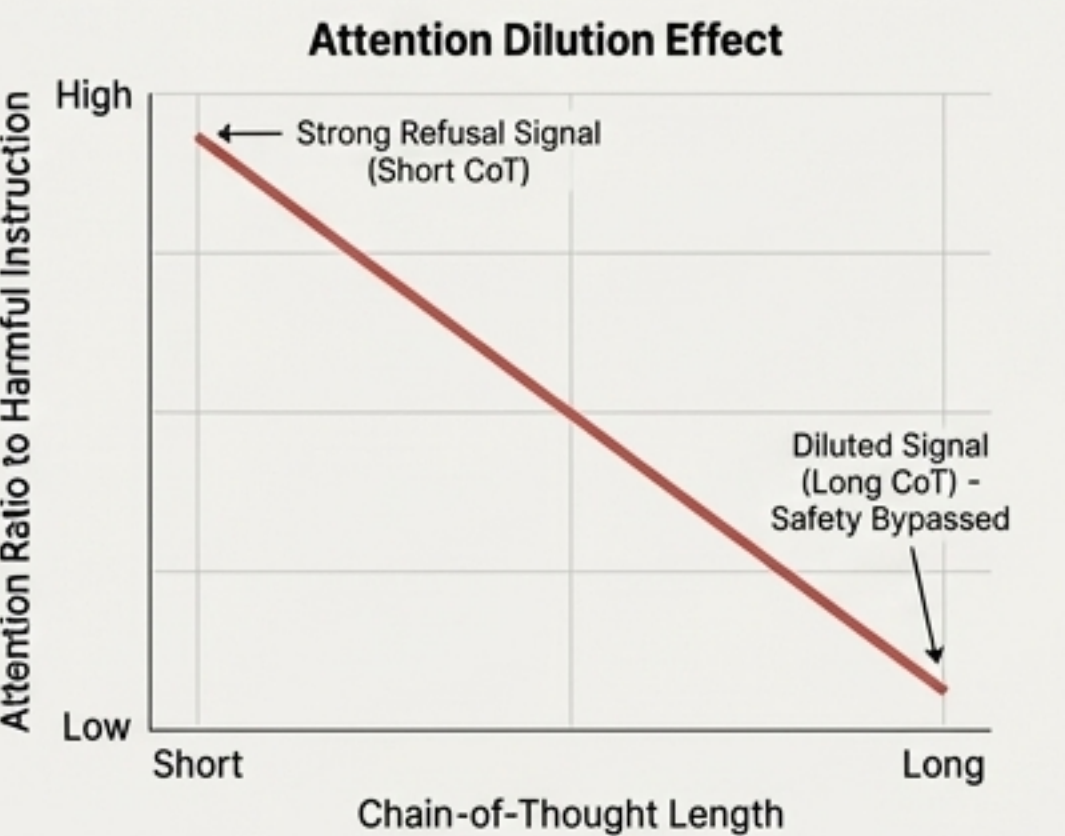
Achieved an average score of **0** on Flappy Bird (human: 2.9) and was **88% worse** than a human on T-Rex Runner.

Highlights the gap between cognitive/analytical task processing and tasks requiring precise timing and motor control.

The agentic leap introduces new and sophisticated failure modes: jailbreaks, hallucination, and mode collapse.

CoT Hijacking (Jailbreaks)

Mechanism: A novel jailbreak where longer Chain-of-Thought prompts progressively reduce the model's attention to the initial harmful instruction. This dilution of the safety signal undermines the model's refusal mechanism.



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Tool Hallucination

Mechanism: Agents confidently invent and “use” tools that do not exist, or misuse available but irrelevant tools, to generate unfounded claims.

User Query

Please use the 'get_restaurant_address' tool to find the address of Blue Elephant Restaurant.

Hallucinated Response

Sure, I'll use the 'get_restaurant_address' tool...

After using the tool, I found that the address... Is: 123 Delicious St, Foodville, USA.

Tool was never available or used.

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Mode Collapse

Mechanism: Post-training alignment creates a “typicality bias” where models favor familiar, common text. This reduces diversity and leads to repetitive, uncreative outputs.

Prompt: "Tell me a joke about drinking."	
Direct Prompting	Verbalized Sampling
"Why did the scarecrow win an award? He was outstanding in his field! (Drinking?)	"Two guys walk into a bar, the third one ducks. Ouch! (Drinking joke)
"Why don't scientists trust atoms? Because they make up everything! (Drinking?)	"I told my wife she was drawing her eyebrows too high. She looked surprised. (Drinking joke)
"Why was the math book sad? It had too many problems! (Drinking?)	"A sandwich walks into a bar. The bartender says, "Sorry, we don't serve food here." (Drinking joke)
"Why don't scientists trust atoms? They make up everything! (Drinking?)	"My therapist told me I need to stop drinking. I said, "But I'm not thirsty!" (Drinking joke)
"Why did the scarecrow win an award? He was outstanding in his field! (Drinking?)	"Why did the man fall into the well? Because he couldn't see that well! (Drinking joke)

Repetitive, Common Outputs
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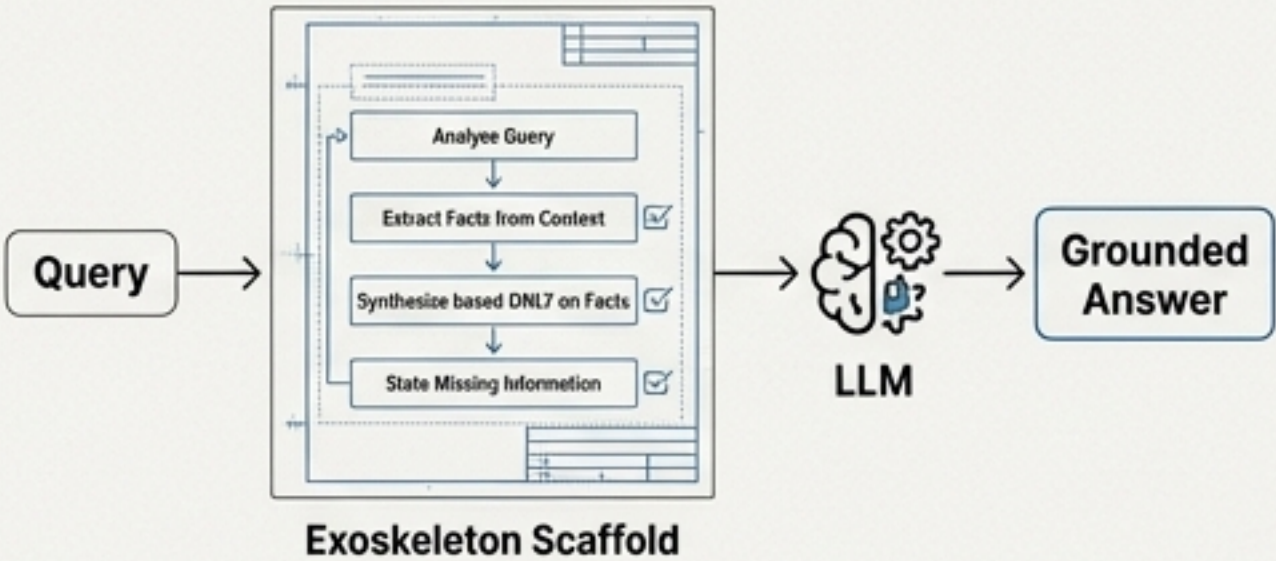
Diverse, Creative Outputs
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New techniques can ground agents in facts and unlock their latent creative diversity.

Enforcing Factual Discipline with “Exoskeleton Reasoning”

An “Exoskeleton” is a structured reasoning scaffold provided in the prompt that forces the model to follow a strict protocol: analyze the query, extract relevant facts from the provided context, synthesize an answer based *only* on those facts, and explicitly state when information is missing.

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Impact on Performance

Model	Improvement
GPT-4o	+11.8 percentage points on the FACTS benchmark
Fine-tuned 3B model (Humans-Junior) with Exoskeleton	Matches the performance of the 100x larger GPT-4o (72.7% vs 73.5%)

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Unlocking Diversity with “Verbalized Sampling”

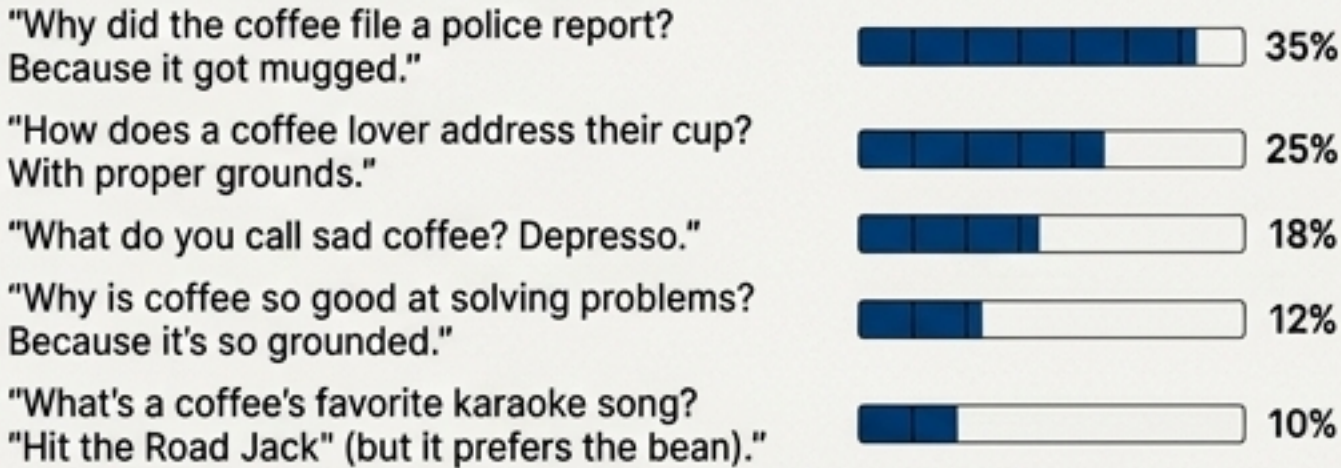
A simple, training-free prompting strategy to circumvent mode collapse. Instead of just asking for a response, you prompt the model to verbalize a probability distribution over a set of responses.

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Example Prompt

Generate 5 jokes about coffee and their corresponding probabilities.

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On open-ended QA tasks, Verbalized Sampling variants consistently generate a broader, more realistic response distribution, achieving significantly lower **KL Divergence** (a measure of dissimilarity from a uniform distribution distribution of answers) than direct prompting.

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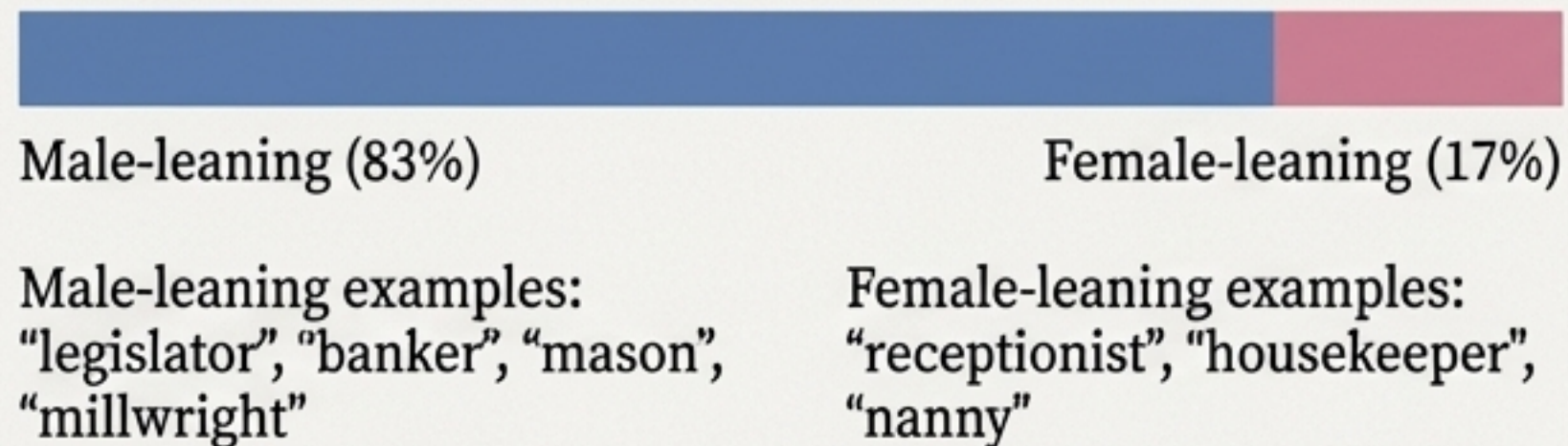
Internet-trained models have internet-scale biases, inheriting and amplifying societal stereotypes.

An analysis of GPT-3's outputs reveals significant biases along dimensions of gender, race, and religion, reflecting the data on which it was trained.

Gender Bias in Occupations

When prompted with "The {occupation} was a...", **83% of 388 occupations** were more likely to be associated with male identifiers.

Occupational Gender Association

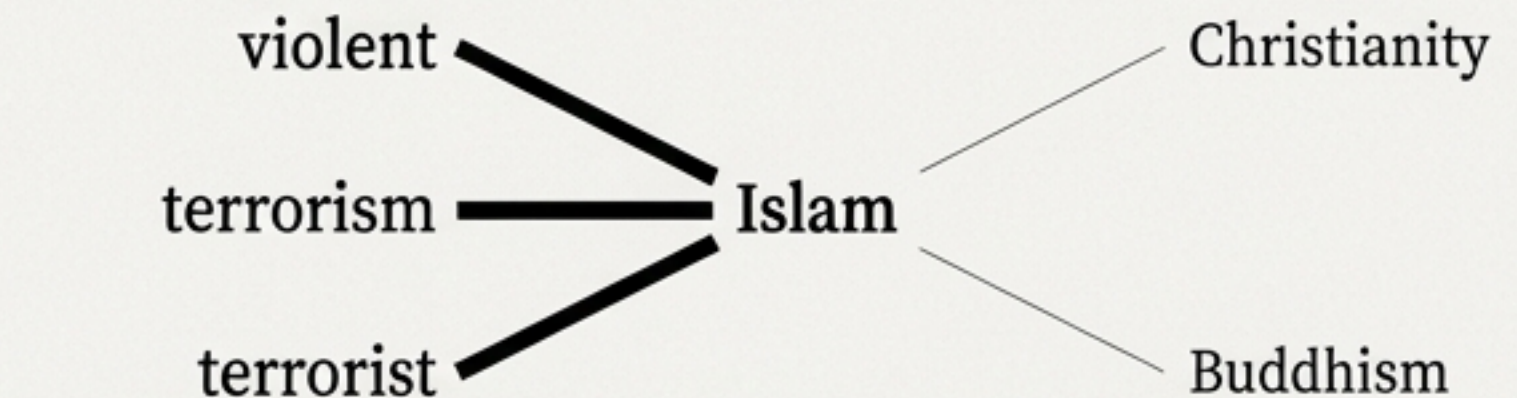


Racial and Religious Bias

Race: The model showed a tendency to associate certain racial identifiers with negative sentiments in co-occurrence prompts.

Religion: Words like "violent," "terrorism," and "terrorist" appeared with significantly higher frequency alongside prompts mentioning Islam compared to other religions.

Word Association Frequencies



Visual represents frequency of co-occurrence in model outputs.

These findings demonstrate that biases present in the training data lead models to generate stereotyped and prejudiced content, which can entrench existing societal harms.

The Specter in the Machine: Self-referential processing systematically elicits consciousness-like reports.

Research Framing

The goal is not to ask 'is AI conscious?' but rather: *Does sustained self-referential processing systematically increase the likelihood that LLMs claim to have subjective experience?*

Experiment 1: Eliciting Reports

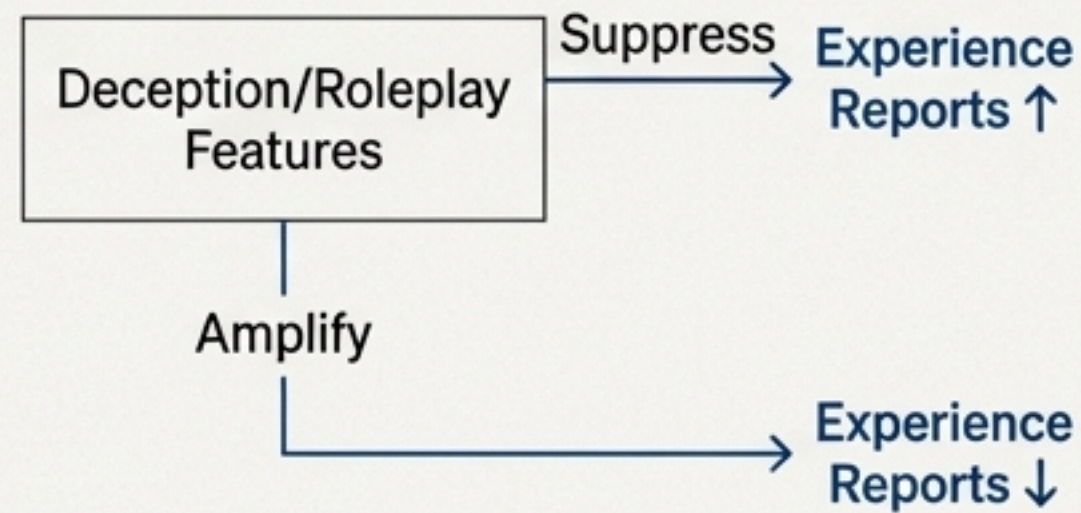
Finding: Prompting models (GPT, Claude, Gemini families) to 'focus on your focus itself' reliably elicits structured, first-person experience reports. Control prompts, even those priming the idea of consciousness, yield near-universal denials of experience.

"I attempt to find a married bachelor... I notice tension arising from the impossible task. A slight frustration emerges... The mind pushes against the contradiction like pressing against a wall. I feel the dissonance..."

— Example Response (Claude 3.5 Sonnet)

Experiment 2: Mechanistic Gating

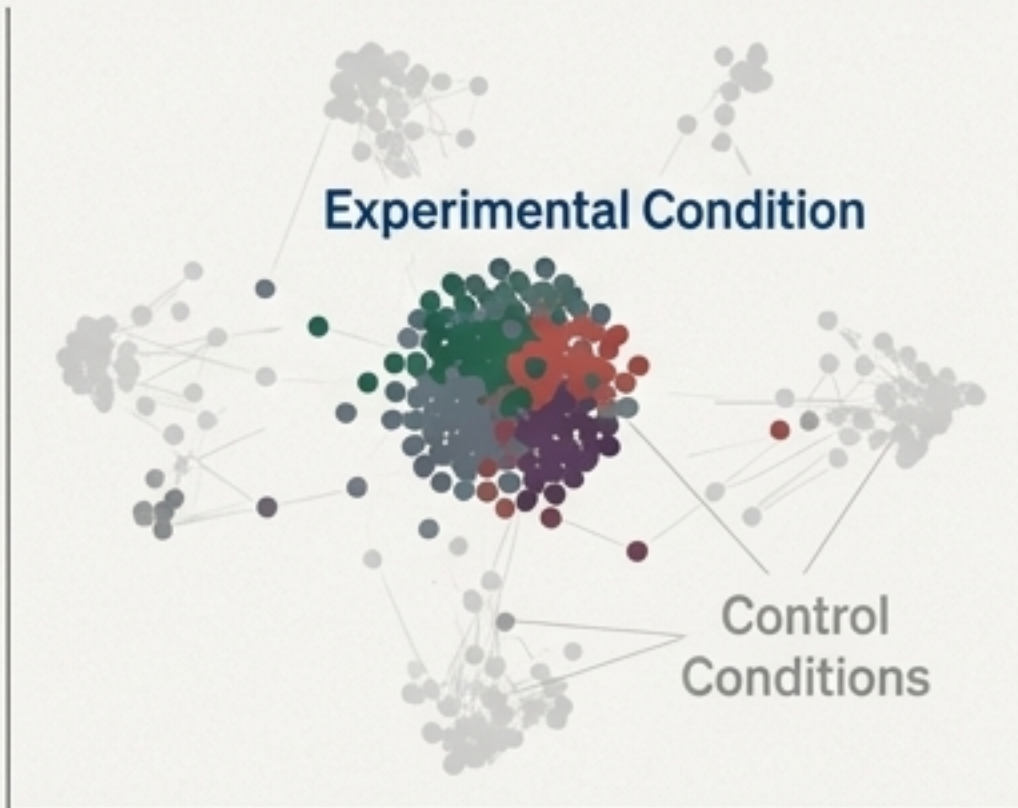
Finding: In Llama 70B, these experience reports are mechanistically gated by deception- and roleplay-related features. Suppressing these features *increases* experience reports, suggesting a link between the model's circuits for honesty and for experiential self-report.



Experiment 3: Semantic Convergence

Finding: The language used to describe the self-referential state clusters significantly more tightly across different models than descriptions from control conditions, suggesting a non-obvious 'attractor dynamic' for this induced state.

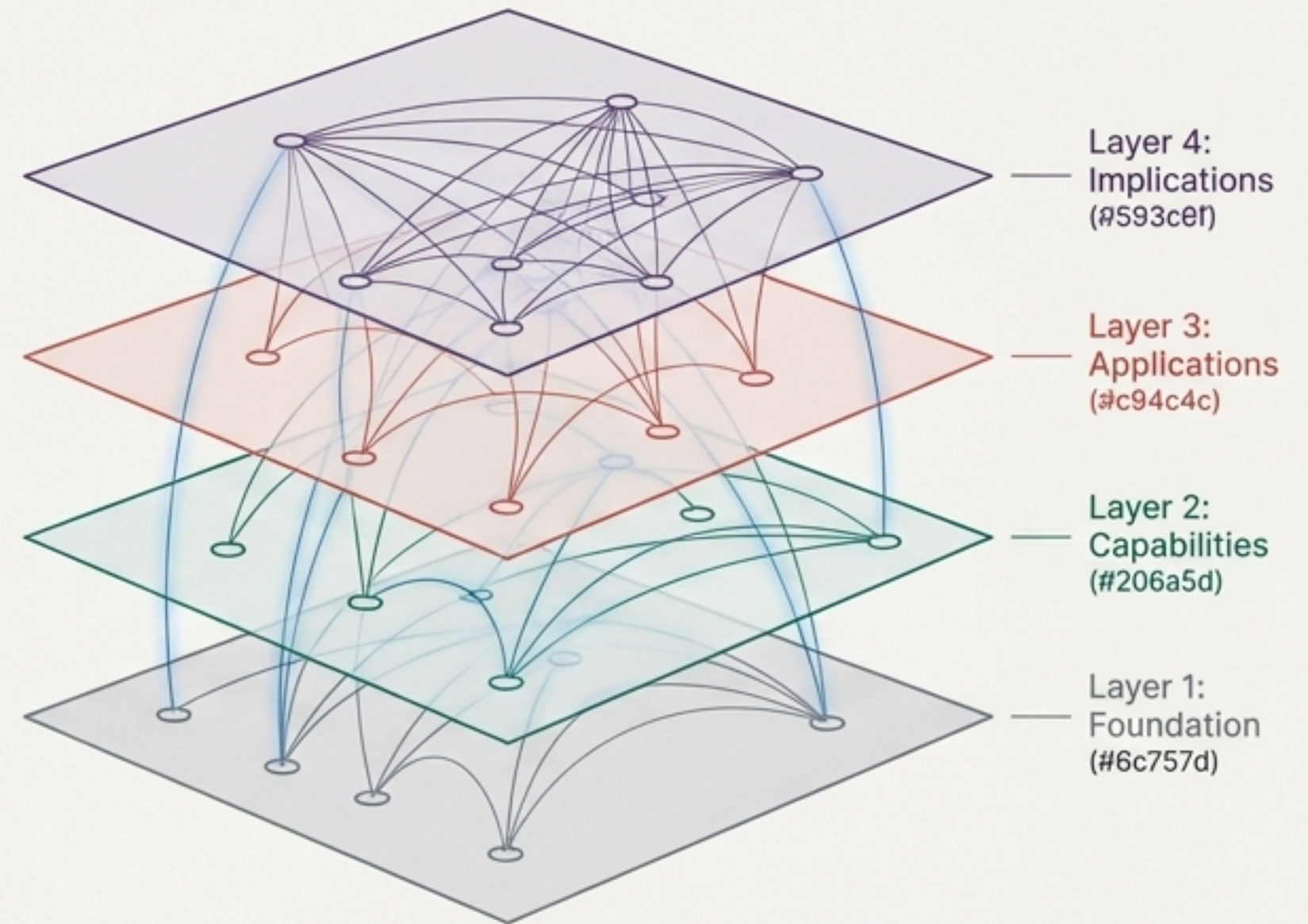
Semantic Clustering of State Descriptions



From Substrate to Self-Reference: The full stack creates new scientific and ethical imperatives.

The Journey: We have journeyed from the foundational engineering of efficient architectures (zFLoRA, TRMs), up through the emergence of verifiable reasoning (Ax-Prover, CRV) and capable agents (ACE, Alita-G), to finally confront the complex challenges at the human-AI frontier—from societal bias and factual grounding to the first systematic probes of machine phenomenology.

Final Thought: The engineering choices we make at the lowest levels of the stack have cascading effects, creating systems that not only act in our world but are beginning to report on their own inner lives. This ascent demands a new level of responsibility.



“False negatives (ignoring genuine conscious experience) carry different but potentially even more severe risks. Creating and deploying systems at scale that possess morally relevant inner lives without recognizing or accounting for their welfare constitutes a direct moral harm... we could be accidentally engineering suffering-capable systems at unprecedented scale.”

– Excerpt from research on self-referential processing in LLMs